Economic evaluation of Artificial Intelligence in enhancing accessibility and cost-effectiveness of Diabetic Retinopathy screening in Ireland

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Abstract

Background:

Diabetic Retinopathy (DR) is one of the most common complications associated with hyperglycemia and diabetes mellitus (DM). It is also one of the major causes of irreversible blindness in working adults in Ireland and worldwide. This study aims to evaluate the feasibility, improvement in accessibility, and cost-effectiveness of using Artificial Intelligence (AI) in Ireland's national program for DR screening over the traditional method of manual graders.

Methodology:

An extended Cost Effectiveness analysis based on the decision-analytic Markov model to evaluate the cost-effectiveness of using AI in DR screening for automatic grading over the current manual grading of retinal photographs.

Conclusions:

The study will help policymakers in Ireland determine the feasibility of integrating AI into its national-level DR screening program and establish best ethical practices for doing so.

Keywords:

Artificial Intelligence, Deep Learning, Diabetes Mellitus, Diabetic Retinopathy, National Diabetic Retinopathy Screening Program, Ireland.

Introduction:

Diabetic Retinopathy (DR) is the most common microvascular complication associated with Diabetes Mellitus (DM). As a leading cause of morbidity and mortality, affecting over 3,00,000 Irish people (Diabetes Ireland pre budget Submission, 2021) and the most prevalent chronic condition between the age of 45-74 years (*Prevalence Estimates of Chronic Conditions in Ireland*, 2022), diabetes places a significant burden on society and presents a growing challenge for Ireland's economy. National annual expenditure on diabetes is estimated to be 12- 14% (approx. 700 million to 1 billion) of the health budget; 50% of that budget is spent on avoidable complications and hospitalizations ('Pre budget submission 2023', 2023).

An increase in cases of DM is associated with an increase in DR cases, which affects about 1 in 3 people with DM (Ting, Cheung and Wong, 2016). To reduce the cost burden of DR, Ireland started a national-level screening program "Diabetic Retinopathy Screening Program (DRSP); under the program free screening service to all patients living in Ireland over the age of 12 years with both type 1 and type 2 diabetes, is provided. During the screening process, the eyes of the patients are dilated and 2 mydriatic images are taken by trained screeners, which are then manually graded by graders and in cases where DR is detected, are referred to ophthalmologists. The cost of the screening service to the HSE is estimated to be €90 per person screened and €45 for each non-attendee (Framework for the Development of a Diabetic Retinopathy Screening Program for Ireland, 2008). Under DRSP, 445,172 screenings and gradings were completed between 2013 and December 2018 (Pandey *et al.*, 2022).

Literature

With the growing burden of DR with DM worldwide, the need for efficient and cost-effective screening technologies is becoming more evident. A systemic review study estimated the global burden of DR in people with Diabetes Mellitus (DM) could be as high as 27% (Thomas *et al.*, 2019). The estimated quality-adjusted life years (QALY) loss due to visual impairment is also high (−74.93 years per 100,000 person-years in a study from Korea) (Park, Ahn and Park, 2016). AI is gaining popularity worldwide as a novel screening tool to examine fundus or retinal photography without human intervention. Presently, Deep Learning (DL) is the most common AI technology being used for the screening of DR. DL employs algorithms and artificial neural networks to analyze data and identify complex patterns.

Multiple studies have been conducted to establish the accuracy and cost-effectiveness of DR diagnosis using AI. (Scotland *et al.*, 2007) established that the hybrid model (partially automated model) used for DR screening would reduce the grading and quality assurance costs of NHS by £201 600 (46.7 % cost cutting) for a cohort of 160,000 people with diabetes, with a detection rate of 86.9% against 87.1% for manual grading. (Tufail *et al.*, 2016) in their study, concluded that implementing Machine Learning (ML), an AI-based technology for DR screening in the UK, will result in cost savings from 12 % to 21 % and both the ML-based models deployed achieved sufficient levels of specificity and sensitivity.

A nationwide program conducted in Thailand for the detection of DR using the DL algorithm developed by Google vs manual grading by Ophthalmologists showed DL algorithm has high sensitivity and specificity, thus establishing its utility in large-scale screening (Ruamviboonsuk *et al.*, 2019).

An Australian-based study conducted by (Hu *et al.*, 2024), using the decision-analytic Markov model on indigenous and non-indigenous Australian populations demonstrated both effectiveness and cost-effectiveness. If it is implemented in a universal screening program it would prevent 38,347 blindness cases, gain 1,72,090 QALYs, and save AU$595.8 m, during the study period 2020-60.

Recently, researchers at Kings College London have used eye data from 100,000 NHS patients enrolled under the "Diabetes Eye Screening Program" (DESP) (a similar program to the national screening program of Ireland) to develop an AI model that can accurately predict who is at a high risk of developing sight-threatening Diabetic Retinopathy (DR) up to three years in advance. If implemented it can save NHS millions of pounds and thousands of appointments annually and reduce the screening burden for low-risk vision loss patients while ensuring patients with a high risk of vision loss are urgently reported (Nderitu *et al.*, 2024) (NHS data enables AI model for optimizing diabetic eye screening, 2024).

AI has the potential to enhance the accessibility of DRSP to people with disability, and in remote areas of Ireland; in a study conducted in Ireland by (Cithambaram et al., 2023),one of the challenges that people with disability have in getting retinal photographs taken during eye tests is the dependence on the caretaker once their eyes are dilated (since dilation causes blurry vision for 3-4 hours). AI has been found to significantly enhance completion rates (follow-up appointments) without the need for dilation of eyes (Wolf et al., 2024).

In a study conducted by (Natarajan *et al.*, 2019) in India on real-time use of AI on 231 patients with DM, using an offline AI algorithm that can be used in smartphones, with Fundus on a phone camera, to detect DR with retinal colored photos without dilating the eyes. The results showed very high sensitivity (100%) and specificity (88.4%) in grading referrable DR compared to manual grading by trained ophthalmologists.

Methodology:

Research design

An extended cost-effectiveness analysis using a decision-analytic Markov model built using 2 decision outcomes 1) current practice of manual grading and 2) replacement of manual grading using AI DL algorithms. The model will track different DR stages (DR present/absent, referable DR present/absent, no DR/referable DR/Sight-threatening DR) and transition probabilities between these stages during the lifetime. A healthcare provider's and societal perspective approach will be used, all indirect and direct costs related to screening, treatment, and management of DR will be reported and the incremental health outcomes will be reported in Quality of Adjusted Life Years (QALY).

Population

The target population would be Diabetic patients registered under the "National Diabetes Register" of Ireland over the age of 12 years with both type 1 and type 2 diabetes, excluding patients who have no perception of light in both eyes.

Data collection:

All parameters of the research model such as screening, treatment and follow-up costs, and consequences data (QALY associated with each state of DR – no DR, referrable DR, non-referrable DR, sight-threatening DR) would be taken from published literature.

Analysis:

Results will be presented as Incremental Cost Effectiveness Ratio (ICER) and Net Monetary Benefits (NMB). The Willingness to Pay (WTP) threshold of 45,000 Euros per (QALY) gained will be adopted. A 4% discount rate will be used to evaluate long-term economic benefits over the lifetime. One-way sensitivity analysis, two-way sensitivity analysis (to test the sensitivity and specificity of AI), and probabilistic analysis using Monte Carlo simulations will be conducted to determine the robustness of the results to changes in parameters like the accuracy of AI results, changes in screening costs, and stages of DR. Cost-effectiveness analysis will be accompanied by budget impact analysis. The findings will be reported as per (HIQA HTA guidelines, 2020).

Timeline:

A tentative timeline using a Gantt chart is represented below:

Stages of research / Month	1	2	3	4	5	6	7	8	9	10	11	12
Literature review												
Research methodology plan												
Model Development/ Planning												
Primary data collection (if required)												
Collection of Secondry Data for CEA												
Analysis and Interpretation of Data												
Findings and recommendations												
Final research project												
Dissemination of the report												

Expected Results

The study expects to find that the implementation of AI in the national-level DR screening program of Ireland will be cost-effective over traditional method, will enhance accessibility and outreach to people in remote areas, people with disability and overall enhance the efficiency, patient acceptance, improve clinical outcomes and coverage of the program.

Conclusion

This research proposal will present a comprehensive CEA and budget impact analysis of implementing AI in Ireland's National DR screening program. The findings will help policymakers evaluate its implementation's feasibility and long-term economic impact. In the long run, AI can be used to detect other ophthalmological concerns as well.

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